Residues and stabilizers in geometric halving

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Abstract

The component construction of geometric halving has a precise residual-connectivity condition: intersecting nonbipartite shadows of an incident flag must not create a bipartite shadow. We give the full residue transport under this condition and compute every maximal stabilizer without assuming residual connectivity of the output. This proves the literal coset identification independently of the stronger hypertope-preservation assertion. We also give an exact criterion for a second leaf and an induced-neighbor coset criterion. The accompanying application treats the regular counterexample and all branches of cubic-toroid halving.

Relation to the common theory and attribution

This paper supplements the complete Topic C common paper [3]; none of its four sections is removed or supplied without its full cost. The construction and source targets are those of Piedade–Tranchida [1]. The residue transport reuses the accepted phase27 proof; the stabilizer and further-leaf sections reproduce their separately reviewed proof packets. The new organization does not claim new authorship of those arguments. Every assertion about a source correction refers to the inspected preprint alone. The conclusions distinguish geometry, residual connectivity, marked intersection property and actual coset identification.

1 The retained cover theorem and its exact residue transport

Throughout this supplement the complete common paper [3] is an acquired dependency. In particular Theorem C.Graph proves the geometry, thinness, natural faithful chamber action, and exact residual-connectivity criterion of the component construction. Its proof includes the connected-truncation lemma, the component-intersection argument and extension of every flag. The following fuller residue identification is the accepted phase27 argument [5]; it is needed to recover residue diagrams rather than connectivity alone.

Let Γ be a residually connected incidence geometry of finite rank at least three, satisfying (B1) and (B2), with connected simple graph Q on types $0, 1$. For a flag F of higher objects write Q_F for its common vertex–edge shadow. Connectedness of these graphs follows by applying the connected-truncation lemma inside the original residues, and chamber extension supplies an edge. Write $D(S)$ for the graph with vertices (v, ϵ) and edges $(v, 0)(w, 1)$ whenever vw is an edge of S. Choose a component C of $D(Q)$. A higher object of the transformed system is (x, C_x), where C_x is a component of $D(Q_x)$ contained in C.

This is literally Construction 3.6 in [1] when Q is nonbipartite: the type-0 part of C_x is an even-walk class, and its type-1 part is the adjacent class, or the same full shadow in the nonbipartite case. If two such components meet at type 1, a neighbor in the connected common graph $Q_{x,y}$ gives a common type-0 point. Thus the stated component incidence is exactly the source incidence, not merely a bijection on its objects. When Q is bipartite, each $D(Q_x)$ has exactly one component inside C; this is its Construction 3.1, attributed there to Lefèvre-Percsy–Percsy–Leemans.

For a higher flag $\Phi = \{(x, C_x) : x \in F\}$, put $W_\Phi = \bigcap_{x \in F} C_x$, with $W_\varnothing = C$. The component lemma of C.Graph states that W_Φ is one component of $D(Q_F)$ if any Q_x is bipartite, and is all of $D(Q_F)$ otherwise. Its connectedness is therefore governed by

$$[Q_x \text{ nonbipartite for every } x \in F] \implies Q_F \text{ nonbipartite.} \tag{PC}$$

The condition ranges over actual higher flags, not arbitrary sets of objects. It is automatic for globally bipartite Q.

Theorem 1 (Full residue transport)**.** *Assume (PC). For a flag of higher lifts Φ, its residue is isomorphic to the component construction on the original residue Γ_F, with selected component W_Φ of $D(Q_F)$. If a defining flag contains a new vertex, its residue is isomorphic to the corresponding original residue. The local component construction again satisfies its parity-compatibility requirement.*

Proof. The vertex case is the explicit isomorphism of C.Graph: the new neighbor of $(v, 0)$ is replaced by its unique original edge at v, and a higher lift is replaced by its original object. Its component is uniquely determined by $(v, 0)$. Simplicity and (B2) preserve incidence. Restrict this map for a larger flag containing that vertex.

For a flag of higher lifts, (PC) makes W_Φ connected. Suppose first that Q_F is bipartite. For any further object y incident with F, its connected shadow $Q_{F,y}$ has exactly one component of its double cover in W_Φ. That component lies in a unique component of $D(Q_y)$. This component meets each prescribed C_x over $Q_{F,y}$, and therefore gives the unique compatible global lift. Conversely a compatible global lift restricts to that component: apply the component-intersection lemma to $F \cup \{y\}$. This establishes the object bijection, including when F is empty.

If Q_F is nonbipartite, every Q_x, $x \in F$, is nonbipartite. For each further y, (PC) on $F \cup \{y\}$ shows that $Q_{F,y}$ is nonbipartite exactly when Q_y is nonbipartite. Indeed the forward implication follows from containment, and the reverse is precisely (PC). Thus restriction from components of $D(Q_y)$ to components of $D(Q_{F,y})$ is a bijection, with either one or two components. Each restriction is compatible with every member of Φ.

In either case, apply the same component-intersection argument to $F \cup \{y, z\}$. Compatible components for an incident pair have nonempty common restriction exactly when their corresponding global lifts are incident. Here a prescribed bipartite constituent selects the same component on every connected subshadow; if all constituents are nonbipartite, (PC) makes the total shadow nonbipartite. This proves preservation of higher incidence. Vertex incidence is membership, so is preserved directly. These maps identify the complete residues.

Finally, in the nonbipartite local case, if every singleton shadow $Q_{F,y}$ of a local flag is nonbipartite, all Q_y and all Q_x for $x \in F$ are nonbipartite. Applying global (PC) to their union makes their total common shadow nonbipartite. The bipartite local case uses the bipartition construction and needs no further condition. □

In a thin rank-two residue, a finite polygon shadow of length m gives a cover component of length m when m is even and $2m$ when m is odd. Its new gonality is therefore $m/2$ or m. For a bi-infinite polygon both components are again bi-infinite paths. The vertex-residue isomorphisms duplicate the adjacent diagram labels. Thus the actual rank-two residues, including infinite ones, are accounted for. Without (PC), the component-intersection description still holds and the transformed system is still a geometry, but the stronger single-component residue assertion is not available.

1.1 Relator parity and the regular-leaf hypothesis

The elementary group calculation $H_J \trianglelefteq U_J = H_J\langle a\rangle$ in C.Endpoint also proves the word form of source Proposition 4.3. For a presentation on a, b, r_i, assign $1 \in \mathbb{F}_2$ to a and 0 to every other marker. This gives a homomorphism from the free group. It factors through the presented group exactly when every relator has even total a-parity, since then the relators and their normal closure lie in its kernel. If it factors, its kernel contains $\langle aba, b, r_i\rangle$ and excludes a, so the index bound identifies that kernel with the halving subgroup. Conversely, if that subgroup is proper, its quotient character has these generator values, and every relator must have even a-parity. The parity lemma below identifies this condition with bipartiteness of the vertex graph. This proves all four equivalences with a fixed presentation; no finite order computation is used.

For completeness, the commuting-marker hypothesis at a regular leaf is supplied by the geometry. Fix all types other than $0, i$, with $i \geq 2$. Its fixed type-1 edge has two endpoints. (B2) says that any compatible

type-i object is incident with both. Thinness makes this rank-two residue a digon. Its two panel involutions therefore commute, giving $[a, r_i] = 1$. This is the exact bridge from (B1)/(B2) and the regular input to the algebra used below.

We give a stabilizer proof of the coset identification in Theorems B and 4.4 of [1]. The proof does not require the halved geometry to be residually connected. This distinction matters: the assertion about residual connectedness in the inspected preprint has counterexamples, whereas the coset identification below remains valid. No claim is made about an uninspected later edition.

2 The construction and the retained geometric facts

Let Γ be a regular hypertope of finite rank $n \geq 3$ with distinguished generators

$$G = \mathrm{Aut}(\Gamma) = \langle a, b, r_2, \dots, r_{n-1} \rangle, \qquad [a, r_i] = 1 \quad (i \geq 2).$$

The generators are involutions and change, respectively, the objects of types $0, 1, 2, \dots, n-1$ in the base chamber $(v, \{v, w\}, x_2, \dots, x_{n-1})$, where $w = av$. Assume that the vertex–edge truncation is a simple graph Q, and that an edge is incident with a higher object exactly when both its endpoints are incident with that object. These are conditions (B1) and (B2) of [1].

For a graph S, its bipartite double cover $D(S)$ has vertices (u, ϵ), with $\epsilon \in \{0, 1\}$, and edges $(u, 0)(z, 1)$ whenever uz is an edge of S. Choose the component C of $D(Q)$ containing $(v, 0)$. For a higher object x, let Q_x be its vertex–edge shadow. The objects of the halved incidence system L are the vertices of C, with their two types, and the pairs (x, C_x), where C_x is a component of $D(Q_x)$ contained in C. Vertex incidence is graph adjacency or membership in C_x; two higher objects are incident when their original objects are incident and their chosen components intersect.

We retain the previously proved component-intersection lemma for this construction. It uses connectedness of the original flag shadows, (B1), and (B2), and does not assume residual connectedness of L. Its consequences needed here are that L is a thin geometry and that its chambers correspond bijectively to the original chambers whose oriented edge has its lift in C. The component of each higher object is uniquely determined by the lifted first endpoint. In particular, when Q is nonbipartite every original chamber occurs; when Q is bipartite exactly the chambers whose first endpoint belongs to the chosen color class occur. The full common proof, including extension of every flag to a chamber, is Theorem C.Graph in the accompanying common paper and the residue transport proved above.

3 An elementary parity lemma

Lemma 2. *Suppose a group $K = \langle a, b, R \rangle$ acts transitively on the vertices and edges of a connected graph S, the stabilizer of a base vertex u is $P = \langle b, R \rangle$, and the edge orbit is that of $\{u, au\}$. Suppose a, b are involutions and a commutes with every member of R. Put $K^+ = \langle aba, b, R \rangle$. Then K^+ is normal in K with index at most two. It is proper exactly when S is bipartite. In that case it is the subgroup preserving the two color classes. Equivalently, it is the stabilizer of either component of $D(S)$ under the action preserving the second coordinate.*

Proof. Conjugation by a exchanges b and aba and fixes R. Thus a normalizes K^+, and $K = K^+\langle a \rangle$. If K^+ is proper, the character $K \longrightarrow K/K^+ \cong C_2$ is constant on cosets of P, since $P \leq K^+$. It colors the vertices gu by the coset of g. The endpoints gu and gau have different colors, so the coloring is a bipartition of S.

Conversely, the bipartition of a connected bipartite graph is unique up to exchange. The action of K on its two classes is a homomorphism to C_2. Every element of P fixes u and therefore preserves the classes, while a exchanges them. Consequently its kernel contains K^+ and excludes a. The index bound gives equality with the kernel. Finally, an even walk stays in a color class and an odd walk changes it. Hence $D(S)$ has two components in the bipartite case, and one otherwise; their stabilizers have exactly the description asserted. □

4 All maximal parabolic stabilizers

Put

$$t_0 = aba, \quad t_1 = b, \quad t_i = r_i \ (i \geq 2), \qquad H = \langle t_0, \dots, t_{n-1} \rangle, \qquad H_i = \langle t_j \ : \ j \neq i \rangle.$$

Theorem 3. *The group H acts regularly on the chambers of L and is its full group of type-preserving automorphisms. In its base chamber, the stabilizer of the object of type i is exactly H_i. Consequently L is isomorphic to the typed coset incidence geometry of H and $(H_i)_{0 \leq i < n}$. None of these assertions requires residual connectedness of L.*

Proof. Apply Lemma 2 first to G and Q. The original vertex stabilizer is $P = \langle b, r_2, \dots, r_{n-1} \rangle$; this is the standard parabolic stabilizer in a regular hypertope. If Q is nonbipartite, then $H = G$. If it is bipartite, H has index two and preserves the chosen class. The chamber correspondence above and the regular action of G on original chambers show that H acts regularly on the chambers of L.

The stabilizer of $(v, 0)$ is $P = H_0$. The stabilizer of $(w, 1)$ is $aPa = \langle aba, r_2, \dots, r_{n-1} \rangle = H_1$. Both subgroups are contained in H, so these are also the stabilizers for the restricted action.

Fix $i \geq 2$ and set

$$K_i = \langle a, b, r_j \ : \ j \geq 2, \ j \neq i \rangle.$$

This is the full stabilizer of x_i in G. In the original residue at x_i, it is transitive on vertices and edges, and its vertex stabilizer is $\langle b, r_j \ : \ j \geq 2, \ j \neq i \rangle$. The shadow graph Q_{x_i} is connected: every truncation on at least two types of a residually connected geometry is connected. Its edges are the translates of the base edge. Thus Lemma 2 applies inside K_i, even if its action on the shadow graph has a kernel. It says that the stabilizer in K_i of the chosen component of $D(Q_{x_i})$ is precisely

$$K_i^+ = \langle aba, b, r_j \ : \ j \geq 2, \ j \neq i \rangle = H_i.$$

An element fixing (x_i, C_i) must fix x_i and preserve C_i. Since $H_i \leq H$, its stabilizer in H is exactly H_i.

The elements t_i carry the base chamber to its unique i-adjacent chamber. For $i \geq 2$ this follows from the old adjacency and the uniqueness of the chosen components through the retained endpoint. The element b fixes v and changes the other endpoint of the base edge, so it gives adjacency of type 1. Conjugating this operation by a gives aba, which fixes w and changes the endpoint of type 0. It preserves every higher component, because it fixes the corresponding higher object and the retained lifted endpoint. These are genuine changes, by thinness of the original panels. Since the t_i generate H and H is chamber-transitive, the chamber graph of L is connected. In a thin geometry, a type-preserving automorphism fixing a chamber fixes each adjacent chamber; induction along galleries shows that it fixes every object. Thus the action of the full automorphism group on chambers is free. The regular subgroup H is already transitive, and hence is the full automorphism group.

Lastly send a left coset hH_i to the image under h of the base object of type i. The stabilizer calculation makes this a bijection on each type. Intersecting cosets give incident objects. Conversely, incident objects extend to a chamber of L, and chamber transitivity supplies an element in the two corresponding cosets. This proves the incidence isomorphism, with distinct types retained even if subgroups coincide. □

The theorem proves the literal coset assertion of [1]; it does not infer the subgroup intersection property from chamber regularity. A stabilizer of a flag of two or more higher objects need not be generated by the remaining t_i. That is precisely where the previously established parity condition and the counterexamples to residual connectedness enter. The present argument supplies a missing proof of a known assertion, not a newly solved open conjecture.

We use the complete parity-cover theorem and its notation. Thus Γ is a residually connected geometry with a simple vertex–edge graph Q on types $0, 1$, satisfying (B2): an edge is incident with a higher object exactly when both its endpoints are. For a higher flag F, Q_F is its common vertex–edge shadow. It is connected and contains an edge. Write $D(Q)$ for the bipartite double cover, and choose a component C. The halved geometry has the vertices of C as its two new types; a higher object is (x, C_x), where C_x is a component of $D(Q_x)$ inside C. Incidence is membership for a lifted vertex and nonempty component

intersection for incident higher objects. The parity-cover theorem proves flag extension without an additional parity hypothesis. It proves residual connectedness precisely under parity compatibility. Those full proofs are retained inputs, not inferred from this note's finite example.

Two further elementary consequences recover the graph and duality clauses of the source. If Q has girth g, the girth of its double cover is the smaller of the least even cycle length in Q and twice the least odd cycle length, omitting a term when that kind of cycle does not exist. Indeed an even cycle lifts to a cycle of its own length, and an odd cycle traversed twice lifts to a cycle of twice its length. Conversely a shortest cover cycle projects to a closed walk without immediate reversals. If the projection is simple, it is an even cycle. If it is not simple, cutting at repeated vertices either exhibits an even cycle no longer than the walk, or exhibits at least two odd cycles, counted with multiplicity. The connecting paths contribute only additional length. This gives the claimed lower bound. Thus, if g is even, the new gonality is $g/2$; if g is odd, it is $\min\{g, k/2\}$, where k is the least even cycle length when one exists. This is source Lemma 3.5.

When Q is nonbipartite, $C = D(Q)$ and the involution $(v, \epsilon) \mapsto (v, 1-\epsilon)$ exchanges the new types. It sends a higher component to its image under the same involution, preserving all incidences. It is the duality of Proposition 3.15.

5 The second leaf

Suppose that two further types k, l also satisfy (B1) and (B2), with type k as vertices. Write the two endpoints of a type-l object y as x_1, x_2. Conditions (B2) at this leaf give

$$Q_y = Q_{x_1} \cap Q_{x_2}$$

as vertex–edge subgraphs, even though $\{x_1, x_2\}$ is not a flag. In particular Q_y is contained in each endpoint shadow.

Theorem 4. *The halved geometry satisfies (B1) and (B2) at (k, l) if and only if every y with bipartite Q_y has at least one endpoint x_j with bipartite Q_{x_j}.*

Proof. Fix a lift (y, C_y). Since C_y is connected and is a subgraph of each $D(Q_{x_j})$, it lies in a unique component C_{x_j}. These are exactly its two lifted endpoints. An endpoint component meeting C_y must be that component; hence there are always exactly two endpoints. Two distinct lifted edges with the same endpoints must project to the same y, by original (B1). Such a pair exists exactly when Q_y is bipartite, giving two components, while both Q_{x_j} are nonbipartite, so both components are contained in the same endpoint lifts. If an endpoint shadow is bipartite, its two components separate the two lifts of y. This proves that the displayed condition is equivalent to (B1).

Assume the condition, and consider (B2). Any object incident with (y, C_y) is incident with both its endpoints: for a lifted vertex this follows from $C_y \subseteq C_{x_j}$; for a higher object it follows from the same inclusion and original (B2).

Conversely, let an object be incident with both endpoints. First let it be a lifted vertex v^ϵ. Original (B2) puts v in Q_y. If Q_y is nonbipartite, its only component contains both lifts of every vertex, so $v^\epsilon \in C_y$. Otherwise choose an endpoint with bipartite shadow. The restriction of its selected component to $D(Q_y)$ is exactly C_y, so the same conclusion follows.

Now let the object be (z, C_z) of a further type. Original (B2) gives $z * y$, so $Q_{y,z}$ is connected and contains an edge. If Q_y is nonbipartite, $C_y = D(Q_y)$; since C_z projects onto every vertex of Q_z, it meets C_y over a vertex of $Q_{y,z}$. Otherwise choose a bipartite endpoint shadow Q_x. The selected components C_z, C_x meet by the given incidence. Their restrictions to the connected graph $Q_{z,x}$ are compatible: C_x selects one bipartite-cover component, and C_z either selects that component or contains both. Consequently their intersection projects onto every vertex of $Q_{z,x}$. Restrict to $Q_{y,z} \subseteq Q_{z,x}$. The component selected by C_x there is the restriction of C_y, so C_z meets C_y. This proves the converse incidence, and hence (B2). □

If an automorphism preserving the types exchanges x_1, x_2 while fixing y, their shadows have the same bipartiteness. In particular this holds in a flag-transitive original geometry. In that case the criterion

reduces to the one printed in Proposition 3.17 of Piedade–Tranchida: along every incidence $x * y$, either both shadows are bipartite or neither is. In general it suffices that one endpoint is bipartite. The necessity argument in the printed proof implicitly identifies the bipartiteness of the two endpoint shadows.

A finite example distinguishing the two criteria

Let $V = \mathbb{Z}/5\mathbb{Z}$, and let Q be its cycle, with edges $e_i = \{i, i+1\}$. Use four types. Types $0, 1$ are $V, E(Q)$. Type 2 consists of X and x_j for $j \in V$; type 3 consists of y_j for $j \in V$. The vertex shadow of X is V, and those of x_j, y_j are $V \setminus \{j\}$. A vertex or edge is incident with one of these objects exactly when it is contained in its shadow. The only incidences between types $2, 3$ are

$$X * y_j, \qquad x_j * y_j.$$

The two truncations $(0, 1)$ and $(2, 3)$ are respectively the cycle and a star with five edges. They satisfy (B1). They also satisfy (B2): the induced subgraph on $V \setminus \{j\}$ is the path $Q - j$, and the intersection of the shadows of X, x_j is that same path.

Every flag extends to a chamber. If a type-3 object, or a type-2 object other than X, is specified, the remaining vertex and edge can be chosen in the nonempty path $Q - j$; a specified vertex has a neighbor in that path. Complete the higher types with y_j and either X or x_j, respecting any prescribed object. If neither is specified, choose any compatible cycle edge and then any j outside its two endpoints. These choices also cover flags already containing an edge.

Here is a complete check of residual connectedness. For a flag of types $0, 1$, the remaining incidence graph is the star at X with its incident edges subdivided by the available y_j. For a flag (v, X), the two incident cycle edges are joined through any y_j with $j \notin \{v-1, v, v+1\}$; every other available y_j is incident with at least one of the two edges. For (v, x_j) the remaining graph is a star at y_j. Flags of types $0, 3$ give $K_{d,2}$ with $d = 1$ or 2; flags of types $1, 2$ give $K_{2,3}$ or $K_{2,1}$; flags of types $1, 3$ give $K_{2,2}$. Finally a flag of types $2, 3$ has the vertex–edge incidence graph of $Q - j$ as its residue. These are all six possible pairs of types.

In a residue at a vertex v, the object X joins all remaining edges, and every remaining x_j, y_j is incident with one of them. In a residue at an edge its two endpoints and X form a connected core to which every remaining object attaches. Residues at X, x_j or y_j have the connected vertex–edge core Q or $Q - j$, and the other objects attach to it. The whole geometry has the core Q. Thus all residues of rank at least two are connected.

Now $Q_X = C_5$ is nonbipartite, whereas every $Q_{x_j} = Q_{y_j}$ is the bipartite path on four vertices. The printed condition fails at each $X * y_j$, but Theorem 4 applies using endpoint x_j. The halved geometry therefore still satisfies both conditions at $(2, 3)$. It is also residually connected: every higher flag other than $\{X\}$ has a bipartite constituent, so the parity condition is automatic. The object counts before and after halving are $(5, 5, 6, 5)$ and $(5, 5, 11, 10)$, respectively. Both geometries have 60 chambers. A separate literal incidence program checks all 322 original and 362 transformed flags. The preceding proof does not rely on that program. The example is not thin or regular; it addresses precisely the general assertion of Proposition 3.17 and the corresponding clause of Theorem 3.18.

6 Stabilizers and the induced neighbor graph

Let Γ now be a regular leaf hypertope, with distinguished involutions a, b, r_i, where a commutes with every r_i. Fix a base chamber, its vertex p, its edge e, and its other endpoint $q = ap$. Put

$$P = G_p = \langle b, r_i \rangle, \qquad R = G_{p,e} = \langle r_i \rangle.$$

All groups and actions below are the actual faithful geometric actions.

Proposition 5. *Condition (B1) holds exactly when $P \cap aPa = R$. If it fails, the marked halving subgroup $\langle aba, b, r_i \rangle$ fails the intersection property.*

Proof. The pointwise stabilizer of p, q is $P \cap aPa$. It acts transitively on the edges joining them. Indeed flag transitivity sends (p, e) to (p, e'); its element fixes p and sends the other endpoint q to itself. The stabilizer of the base edge in this action is R. Thus the set of edges with these endpoints is in bijection with $(P \cap aPa)/R$. Every edge has two distinct endpoints by thinness and the leaf condition, so (B1) is equivalent to this index being one. All pairs of adjacent vertices are equivalent under the group, so the base pair suffices.

For the new marking its two endpoint omission subgroups are $H_0 = P$ and $H_1 = aPa$, while $H_{0,1} = R$. A strict intersection therefore violates the marked intersection property. □

For a base object x_i of a higher type put $K_i = G_{x_i}$ and $P_i = P \cap K_i$. Identify vertices with left cosets gP, so the neighbors of $p = P$ form PaP/P. This convention is fixed throughout the next proposition.

Proposition 6. *Assume (B1). Condition (B2) is equivalent to*

$$K_i P \cap PaP = P_i aP \qquad (i \neq 0, 1).$$

The literal sufficient condition in source Lemma 4.6,

$$Pa \cap wP = \varnothing \quad (w \in K_i,\ wP \neq aP),$$

implies (B2), but is in fact impossible for a regular leaf hypertope of rank at least three satisfying (B1).

Proof. The vertices incident with x_i are $K_i P/P$. Since $a \in K_i$, the edges at p incident with x_i have their other endpoints $P_i aP/P$: the group P_i is transitive on the appropriate vertex–edge–object flags. Original (B1) identifies an edge uniquely with its endpoints. The displayed equality therefore says exactly that every global neighbor of p in the vertex shadow of x_i is a neighbor within its residue. This is (B2) at p, x_i; flag transitivity gives it at every vertex–object incidence. The forward implication of (B2) also follows directly from the leaf digons.

The printed condition forces any coset in the left-hand set to be aP: an intersection with PaP can be translated on the right within P to give an intersection with Pa. Hence it is a sufficient, stronger condition. To see why it cannot hold here, take $w = ba \in K_i$, which is allowed for any higher type i. Then $ba \in Pa \cap baP$. Moreover $baP \neq aP$. Otherwise b would fix both p, q, and (B1) would force $b \in R$ by Proposition 5, contrary to the original intersection property. Thus this w contradicts the printed condition. The coset equality above gives the nonvacuous criterion. □

7 The two elementary examples

The regular polytope of type $\{3, 3, 2\}$ has group $S_4 \times C_2$, with the three adjacent transpositions on four letters and an independent central involution as its generators. Its vertices, edges and triangles are those of a tetrahedron, and it has two distinct facets, each incident with all those faces. The intersection of the two facets has several maximal triangles and no greatest proper face, so its face poset is not a lattice. Nevertheless the vertex–edge graph is K_4, and incidence with a triangle or either facet is exactly containment of the endpoints. Hence (B1) and (B2) hold. Higher flags are chains, whose common vertex shadow is the smallest shadow, so parity compatibility holds and the repaired halving theorem applies. This proves the assertion of source Example 4.8 without assuming that its face poset is a lattice.

For Example 4.9, realize the hemicube by identifying antipodal faces of the ordinary cube. It has four vertices and six edges, with graph K_4, so (B1) holds. Each of its three faces is a four-cycle using all four vertices but only four of the six edges. An omitted edge has both endpoints on that face, proving failure of (B2). Dually, each pair of its three faces has two common edges, so the opposite leaf fails (B1).

The automorphism group is the signed permutation group in dimension three modulo its central sign change, of order 24, acting faithfully as S_4 on the four antipodal vertex pairs. Choose the usual markers $a = z_1, b = s_{12}, c = s_{23}$: a acts as a double transposition and b, c as transpositions, with orders $|ab| = 4$, $|bc| = 3$ and $[a, c] = 1$. More explicitly, choose the representatives

$$A = (1, 1, 1),\ B = (1, -1, -1),\ C = (-1, 1, -1),\ D = (-1, -1, 1)$$

of the four antipodal pairs. Then $a = (AB)(CD)$, $b = (BC)$, $c = (CD)$ and $aba = (AD)$. The new markers aba, c, b are transpositions. Their successive products have order three and the two endpoints commute. The subgroup $\langle b, c\rangle$ is S_3 fixing one vertex, while aba joins that vertex to another. They are therefore the adjacent transpositions of a path on four vertices and generate S_4 with its simplex marking. Their coset geometry is the tetrahedron. This is an algebraic halving statement; it does not apply the geometric theorem after its (B2) hypothesis has failed.

8 Attribution and limits

The definitions and targets are those of Piedade–Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, especially Propositions 3.17, 4.7, Lemmas 4.5–4.6 and Examples 4.8–4.9. The stabilizer result and examples are their conclusions. The endpoint criterion and the explicit coset equality provide the corrections and reusable arguments supplied here. The fixed preprint's general second-leaf necessity assertion is refuted; its regular flag-transitive specialization remains valid. These are source corrections, not solutions of named open conjectures. No statement is made about an uninspected journal revision, publication priority or Lean verification. The general and regular hypotheses are kept separate throughout.

References

[1] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1 (29 May 2024). Only this full text is in the present source contract.

[2] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Theorem 6D4 and the standard affine Coxeter realization of cubic tessellations.

[3] Human Compression Project, *Changes of chamber generators and binary translation groups*, accepted Topic C common paper, v2. The whole paper, including its binary-module sections, is retained.

[4] Human Compression Project, *Residues and stabilizers in geometric halving*, accompanying common supplement.

[5] Human Compression Project, *Geometric halving through graph covers and parity-compatible shadows*, phase27 accepted proof, 10 October 2026; its complete geometric proof is retained in the old common paper and the present residue supplement.

[6] Human Compression Project, *Stabilizers in geometric halving*, accepted proof reproduced in the accompanying common supplement.

[7] Human Compression Project, *Affine groups after one and two halvings*, accepted proof reproduced in the affine sections of this application.

[8] Human Compression Project, *An actual regular leaf hypertope whose halving loses residual connectivity*, accepted phase30 proof and finite certificate; all seven sections are reworked in this application.